

Homework 1

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Problem 1

(a)

$$2u = [4, -2, 6]^T$$

$$2u - v = [4, -2, 6]^T - [1, 4, -2]^T = [3, -6, 8]^T$$

$$(2u - v) \cdot w = (3)(-4) + (-6)(2) + (8)(6) = -12 - 12 + 48 = 26$$

=> ANS: 26

(b)

$$(1/2)w = [-2, 1, 3]^T$$

$$v + (1/2)w = [-1, 5, 1]^T$$

$$||[-1, 5, 1]^T|| = \sqrt{1 + 25 + 1} = \sqrt{27} = 3\sqrt{3} \approx 5.20$$

=> ANS: $3\sqrt{3} \approx 5.20$

(c)

$$\cos \theta = (u \cdot w) / (||u|| ||w||)$$

$$u \cdot w = -8 - 2 + 18 = 8$$

$$||u|| = \sqrt{4 + 1 + 9} = \sqrt{14}, \quad ||w|| = \sqrt{16 + 4 + 36} = \sqrt{56} = 2\sqrt{14}$$

$$||u|| ||w|| = 2(14) = 28$$

$$\cos \theta = 8/28 = 2/7 \Rightarrow \theta = \arccos(2/7)$$

=> ANS: $\theta \approx 73.40^\circ$

Problem 2

(a)

AB: undefined. A is 3×3 , B is 2×3 , 3 cols $\neq$ 2 rows.

BA: $(2 \times 3)(3 \times 3) = 2 \times 3$.

$$r1 = [1(2) + 0(1) + (-2)(0), 1(-1) + 0(3) + (-2)(4), 1(0) + 0(-2) + (-2)(1)] = [2, -9, -2]$$

$$r2 = [3(2) + 1(1) + 1(0), 3(-1) + 1(3) + 1(4), 3(0) + 1(-2) + 1(1)] = [7, 4, -1]$$

$$\Rightarrow BA = \begin{bmatrix} 2 & -9 & -2 \\ 7 & 4 & -1 \end{bmatrix}$$

(b)

AV: $(3 \times 3)(3 \times 1) = 3 \times 1$.

$$AV = [2(1) + (-1)(-2) + 0(3), 1(1) + 3(-2) + (-2)(3), 0(1) + 4(-2) + 1(3)]^T = [4, -11, 5]^T$$

VA: undefined. V is 3×1 , 1 col $\neq$ 3 rows of A.

$$\Rightarrow \text{ANS: } AV = [4, -11, 5]^T, VA \text{ undefined}$$

(c)

Aw: undefined. A has 3 cols, w is 1×3 so 1 row, $3 \neq 1$.

WA: $(1 \times 3)(3 \times 3) = 1 \times 3$.

$$WA = [2(2) + 0(1) + (-1)(0), 2(-1) + 0(3) + (-1)(4), 2(0) + 0(-2) + (-1)(1)] = [4, -6, -1]$$

$$\Rightarrow \text{ANS: } WA = [4, -6, -1], Aw \text{ undefined}$$

(d)

vw: $(3 \times 1)(1 \times 3) = 3 \times 3$.

$$vw = \begin{bmatrix} 1(2) & 1(0) & 1(-1) \end{bmatrix} \begin{bmatrix} 2 & 0 & -1 \\ -2(2) & -2(0) & -2(-1) \end{bmatrix} = \begin{bmatrix} -4 & 0 & 2 \\ 3(2) & 3(0) & 3(-1) \end{bmatrix} \begin{bmatrix} 6 & 0 & -3 \end{bmatrix}$$

$$vw: (1 \times 3)(3 \times 1) = 1 \times 1 = 2(1) + 0(-2) + (-1)(3) = -1$$

$$\Rightarrow \text{ANS: } vw \text{ above, } vw = [-1]$$

(e)

$$A^2 = AA, (3 \times 3)(3 \times 3).$$

$$r_1 = [2(2) + (-1)(1) + 0(0), 2(-1) + (-1)(3) + 0(4), 2(0) + (-1)(-2) + 0(1)] = [3, -5, 2]$$

$$r_2 = [1(2) + 3(1) + (-2)(0), 1(-1) + 3(3) + (-2)(4), 1(0) + 3(-2) + (-2)(1)] = [5, 0, -8]$$

$$r_3 = [0(2) + 4(1) + 1(0), 0(-1) + 4(3) + 1(4), 0(0) + 4(-2) + 1(1)] = [4, 12, -7]$$

$$\Rightarrow A^2 = \begin{bmatrix} 3 & -5 & 2 \\ 5 & 0 & -8 \\ 4 & 12 & -7 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 0 & -8 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 12 & -7 \end{bmatrix}$$

(f)

B^2 : undefined. B is 2×3 , not square, BB needs 3 cols = 2 rows, false.

Problem 3

(a)

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix} R_2 \rightarrow R_2 - 2R_1 \quad \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix} R_2 \rightarrow -R_2 \quad \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & 1 & | & 2 \end{bmatrix} R_3 \rightarrow R_3 + R_1 \quad \begin{bmatrix} 0 & -1 & 3 & | & 4 \end{bmatrix} R_3 \rightarrow R_3 - 3R_2 \quad \begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & 2 & | & -1 \end{bmatrix} \xrightarrow{\quad} \begin{bmatrix} 0 & 3 & 1 & | & 0 \end{bmatrix} \xrightarrow{\quad} \begin{bmatrix} 0 & 0 & 10 & | & 12 \end{bmatrix}$$

$R_3 \rightarrow (1/10)R_3$:

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 & | & 6/5 \end{bmatrix}$$

$$x_3 = 6/5; x_2 = -4 + 3(6/5) = -2/5; x_1 = -1 - 2(-2/5) + 6/5 = 1$$

$$\Rightarrow \text{ANS: } x = [1, -2/5, 6/5]^T$$

(b)

$$\begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix} R_2 \rightarrow R_2 - 2R_1 \quad \begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix} R_3 \rightarrow R_3 - R_2 \quad \begin{bmatrix} 1 & 0 & -7 & | & 8 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 5 & 1 & | & 6 \end{bmatrix} R_3 \rightarrow R_3 - 3R_1 \quad \begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix} R_1 \rightarrow R_1 - 2R_2 \quad \begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 7 & 0 & | & 10 \end{bmatrix} \xrightarrow{\quad} \begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix} \xrightarrow{\quad} \begin{bmatrix} 0 & 0 & 0 & | & 0 \end{bmatrix}$$

No pivot in col 3 $\Rightarrow z$ free, $z = t$.

$y = -2 - 3t$, $x = 8 + 7t$. Infinitely many solutions.

$\Rightarrow \text{ANS: } [x, y, z]^T = [8, -2, 0]^T + t [7, -3, 1]^T, t \in \mathbb{R}$

Problem 4

(a)

$x_1 = \text{kg of Alloy 1}$, $x_2 = \text{kg of Alloy 2}$, $x_3 = \text{kg of Alloy 3}$ (masses in kg).

Mass of each metal in = mass of that metal out.

$$\text{Cu: } 0.5x_1 + 0.2x_2 + 0.4x_3 = 0.36$$

$$\text{Zn: } 0.3x_1 + 0.7x_2 + 0.2x_3 = 0.44$$

$$\text{Ni: } 0.2x_1 + 0.1x_2 + 0.4x_3 = 0.20$$

(b)

Metal masses on the right: 36, 44, 20 kg. $\times 10$ to clear decimals:

$$[5 \ 2 \ 4 \mid 360]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

(c)

$$R_1 \rightarrow R_1 - 2R_3:$$

$$[1 \ 0 \ -4 \mid -40]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

$$R_2 \rightarrow R_2 - 3R_1, \ R_3 \rightarrow R_3 - 2R_1:$$

$$[1 \ 0 \ -4 \mid -40]$$

$$[0 \ 7 \ 14 \mid 520]$$

$$[0 \ 1 \ 12 \mid 280]$$

$$R_2 \rightarrow (1/7)R_2, \ R_3 \rightarrow R_3 - R_2:$$

$$[1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 520/7]$$

$$[0 \ 0 \ 10 \mid 1440/7]$$

$R_3 \rightarrow (1/10)R_3$:

$$\begin{bmatrix} 1 & 0 & -4 & | & -40 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 2 & | & 520/7 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 & | & 144/7 \end{bmatrix}$$

$$x_3 = 144/7 \approx 20.57$$

$$x_2 = 520/7 - 2(144/7) = 232/7 \approx 33.14$$

$$x_1 = -40 + 4(144/7) = 296/7 \approx 42.29$$

$\Rightarrow$ ANS: Alloy 1 ≈ 42.29 kg, Alloy 2 ≈ 33.14 kg, Alloy 3 ≈ 20.57 kg

Problem 5

(a)

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 6 & 4 \end{bmatrix}$$

$R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 3R_1$:

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$R_3 \rightarrow R_3 - R_2, R_1 \rightarrow R_1 - R_2$:

$$\begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$\Rightarrow \text{RREF}(A) = \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

(b)

2 pivots in the RREF (cols 1 and 3) $\Rightarrow \text{rank}(A) = 2$

(c)

Same ops on $[A \mid b]$ leave row 3 as $[0 \ 0 \ 0 \mid b_3 - b_1 - b_2]$.

0 solutions: possible. $b_3 \neq b_1 + b_2$, e.g. $b = [0, 0, 1]^T$, gives row 3: $0 = 1$, inconsistent.

1 solution: impossible. Col 2 is never a pivot col, so x_2 is free whenever the system is consistent; rank $2 < 3$ unknowns, so never unique.

Infinitely many: possible. $b_3 = b_1 + b_2$, e.g. $b = [1, 3, 4]^T$, is consistent and x_2 free $\Rightarrow$ infinitely many.

Problem 6

(a)

Vectors as columns:

$$\begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_1:$$

$$\begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2:$$

$$\begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 2 \end{bmatrix}$$

3 pivots for 3 vectors $\Rightarrow$ only $c_1 = c_2 = c_3 = 0$ solves $c_1v_1 + c_2v_2 + c_3v_3 = 0$.

$\Rightarrow$ ANS: independent, span = all of $\mathbb{R}^3$

(b)

$$\begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$R_2 \rightarrow R_2 - R_1:$

$$\begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$R_3 \rightarrow R_3 - R_2:$

$$\begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 2 \end{bmatrix}$$

3 pivots for 3 vectors $\Rightarrow$ independent, and 3 independent vectors in $\mathbb{R}^3$ span $\mathbb{R}^3$.

$\Rightarrow$ ANS: independent, span = all of $\mathbb{R}^3$